

GRAVITATION

Complete Concepts • Formulae • Examples • Practice Questions • Solutions

Chapter Snapshot

This chapter explains the force that attracts all objects toward each other. You will learn universal gravitation, free fall, acceleration due to gravity, mass and weight, thrust and pressure, buoyancy, Archimedes' principle, and relative density.

1. Gravitation

Gravitation is the force of attraction between any two objects having mass. For example, the Earth attracts us, the Moon attracts the Earth, and the Sun attracts planets.

Newton's Universal Law of Gravitation

Every object in the universe attracts every other object with a force directly proportional to the product of their masses and inversely proportional to the square of the distance between them.

$$F = G \frac{m_1 m_2}{r^2}$$

F = gravitational force (N) m_1, m_2 = masses (kg) r = distance between centres (m) G = universal gravitational constant

Value of $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$

Important

Gravitational force is always **attractive**. If distance becomes $2r$, force becomes $F/4$. If one mass becomes double, force becomes $2F$.

Importance of Gravitation

- Keeps us bound to Earth
- Holds the atmosphere around Earth
- Causes the Moon to revolve around Earth
- Keeps planets in their orbits
- Produces ocean tides

2. Free Fall

When an object falls toward Earth under the influence of gravity alone, its motion is called **free fall**. Air resistance is ignored in the ideal case.

During free fall, acceleration is called **acceleration due to gravity (g)**.

$$g \approx 9.8 \text{ m/s}^2$$

Near Earth's surface, g is approximately constant and acts toward the centre of Earth.

Equations of Motion for Free Fall

For downward motion, take acceleration = $+g$ if downward is chosen positive:

$$v = u + gt$$

$$s = ut + \frac{1}{2}gt^2$$

$$v^2 - u^2 = 2gs$$

For upward motion, gravity acts opposite to upward motion, so acceleration is $-g$.

3. Mass and Weight

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Amount of matter in a body	Gravitational force acting on a body
SI unit: kilogram (kg)	SI unit: newton (N)
Constant everywhere	Changes when g changes
Scalar quantity	Force; has direction toward Earth
Measured by beam balance	Measured by spring balance

$$\text{Weight} = \text{mass} \times \text{acceleration due to gravity} = mg$$

Moon Fact

The Moon's gravity is about 1/6 of Earth's gravity. Therefore a person's **mass remains the same**, but their **weight decreases** on the Moon.

4. Thrust and Pressure

Thrust is the force acting perpendicular to a surface. **Pressure** is thrust acting per unit area.

$$\text{Pressure} = \text{Thrust} / \text{Area}$$

SI unit of pressure = pascal (Pa) = N/m^2

Everyday Applications

A sharp knife cuts easily because its small area produces high pressure. School bags have broad straps to reduce pressure on shoulders. Dams are thicker at the bottom because liquid pressure increases with depth.

5. Buoyancy

When an object is immersed in a fluid, the fluid exerts an upward force on it. This upward force is called **buoyant force** or **upthrust**.

An object may float, sink, or remain suspended depending on the relation between its weight, buoyant force, and density.

Archimedes' Principle

Statement

When an object is immersed partially or completely in a fluid, it experiences an upward force equal to the **weight of the fluid displaced** by the object.

Applications: ships and submarines, hydrometers, lactometers, finding purity of substances, and determining relative density.

6. Density and Relative Density

$$\text{Density} = \text{Mass} / \text{Volume}$$

SI unit of density = kg/m³

$$\text{Relative Density} = \text{Density of substance} / \text{Density of water}$$

Relative density has **no unit** because it is a ratio of two similar quantities.

Floating Rule

If average density of an object is less than the density of the fluid, it tends to float. If it is greater, it tends to sink.

Solved Examples

Example 1: Gravitational force

Two objects of masses 10 kg and 20 kg are 2 m apart. Find the gravitational force.

$$F = Gm_1m_2/r^2 = (6.67 \times 10^{-11} \times 10 \times 20)/4$$

$$F = 3.335 \times 10^{-10} \text{ N}$$

Example 2: Weight

Find the weight of a 50 kg student on Earth. Take $g = 9.8 \text{ m/s}^2$.

$$W = mg = 50 \times 9.8 = 490 \text{ N}$$

Example 3: Free fall

A stone is dropped from rest for 3 s. Find its final velocity. Take $g = 9.8 \text{ m/s}^2$.

$$v = u + gt = 0 + 9.8 \times 3 = 29.4 \text{ m/s downward}$$

Example 4: Distance fallen

A ball is dropped from rest for 2 s. Find the distance fallen.

$$s = ut + \frac{1}{2}gt^2 = 0 + \frac{1}{2} \times 9.8 \times 4 = 19.6 \text{ m}$$

Example 5: Pressure

A force of 100 N acts on an area of 0.5 m². Find pressure.

$$P = F/A = 100/0.5 = 200 \text{ Pa}$$

Example 6: Density

A block has mass 200 kg and volume 0.1 m³. Find density.

Density = $m/V = 200/0.1 = 2000 \text{ kg/m}^3$

Practice Questions

1. State Newton's universal law of gravitation.
2. What happens to gravitational force if the distance between two bodies becomes three times?
3. Define free fall and acceleration due to gravity.
4. Differentiate between mass and weight.
5. Calculate the weight of a 60 kg person on Earth ($g = 9.8 \text{ m/s}^2$).
6. A body falls freely from rest for 4 s. Find its final velocity.
7. A stone is thrown vertically upward. What happens to its velocity at the highest point?
8. Define thrust and pressure. Why do sharp objects have pointed edges?
9. State Archimedes' principle.
10. Why do objects feel lighter when immersed in water?
11. Define density and relative density. Why does relative density have no unit?
12. A force of 500 N acts on an area of 2 m^2 . Find pressure.
13. A substance has mass 500 g and volume 250 cm^3 . Find density in g/cm^3 .
14. A 5 kg object is on the Moon where $g = 1.63 \text{ m/s}^2$. Find its weight.
15. Explain why a ship made of steel can float on water.

Answers & Solutions

1. Every two objects attract each other with a force proportional to the product of their masses and inversely proportional to the square of the distance between them.
2. $F \propto 1/r^2$. If r becomes $3r$, F becomes $F/9$.
3. Free fall is motion under gravity alone. g is the acceleration produced by Earth's gravitational pull, approximately 9.8 m/s^2 .
4. Mass is amount of matter and remains constant; weight is gravitational force $W = mg$ and changes with g .
5. $W = mg = 60 \times 9.8 = 588 \text{ N}$.
6. $v = u + gt = 0 + 9.8 \times 4 = 39.2 \text{ m/s downward}$.
7. At the highest point, instantaneous velocity becomes **zero**, while acceleration due to gravity still acts downward.
8. Pressure = thrust/area. A pointed edge has smaller area, producing greater pressure for the same force.
9. An immersed object experiences an upward buoyant force equal to the weight of the fluid displaced by it.
10. Water exerts an upward buoyant force, reducing the object's apparent weight.
11. Density = mass/volume. Relative density is density of substance divided by density of water; units cancel.
12. $P = F/A = 500/2 = 250 \text{ Pa}$.
13. Density = $500/250 = 2 \text{ g/cm}^3$.
14. $W = mg = 5 \times 1.63 = 8.15 \text{ N}$.
15. A ship's hollow shape gives it a large volume and low average density. It displaces enough water to produce buoyant force equal to its weight.

Exam Quick Revision

Universal gravitation	$F = Gm_1m_2/r^2$
Acceleration due to gravity	$g \approx 9.8 \text{ m/s}^2$
Free fall equations	$v = u + gt$; $s = ut + \frac{1}{2}gt^2$; $v^2 - u^2 = 2gs$
Weight	$W = mg$
Pressure	$P = F/A$
Density	$\rho = m/V$
Relative density	Density of substance / Density of water
Archimedes' principle	Buoyant force = weight of displaced fluid

Study Tip: Learn the concept first, then the formula, then solve numericals by writing Given → Formula → Substitution → Answer with unit.